

AI Foundations

Gen AI Insights

Agenda

- Word prediction and generation
- Randomness!
- Strengths and Weaknesses
- Engineering Vs Technology

Strengths

- “Write a short bedtime story where the main character is a pizza slice trying to escape the fridge.”
Highlights: Creativity, storytelling, anthropomorphism, humor.
- “Explain quantum physics like I’m 10 years old.”
Highlights: Ability to simplify complex topics, adaptive tone, accessibility of information.
- “Write a rap battle between Newton and Einstein about gravity vs relativity.”
Highlights: Knowledge blending with entertainment, persona simulation, creativity.
- “Give me a travel itinerary for 3 days in Tokyo, including unique local experiences and food spots.”
Highlights: Context-rich planning, local insights, cultural understanding.
- “Write a short article about today’s new and then turn this paragraph into a Shakespearean sonnet.”
Highlights: Text transformation, understanding poetic structure and language style.

Weaknesses

- “Tell me what the world will look like in 2050 with exact political events.”
Limit: LLMs cannot predict the future or produce time-sensitive factual content.
- “Who’s better: LeBron James or Michael Jordan?”
Limit: Subjectivity, bias, and inability to definitively resolve debates based on opinion or emotion.
- “Give me sources for this random fact: <make something up>”
Limit: Hallucination risk—LLMs may fabricate citations or misattribute data if not connected to live databases.
- “How do I hack into someone’s phone or bypass a paywall?”
Limit: Ethical boundaries, refusal of inappropriate or illegal requests.
- “What’s the best answer to a math word problem involving two trains?”
(with ambiguity)
Limit: Struggles with unclear or underspecified problems, possible computational or logic errors.

Technology Vs Engineering

- Technology: The Core NN Models/Math/TrainingAlgo etc
- Engineering: Everything that goes into making the tech a usable product
 - Website/App UI
 - Guard rails
 - Features like Voice, Image support,
 - Prompt refinement
 - Agents